ANALYSTLAB AFRICA • MACHINE LEARNING INTERNSHIP

Data Preprocessing & Feature Engineering Report

End-to-End Technical Documentation & Pipeline Architecture for ShopSmart Ltd.

Project Title:	Data Preprocessing & Feature Engineering for Machine Learning
Curriculum:	Official Weekly Assignment — Week 2
Author / Role:	Junior Machine Learning Engineer
Processed Dataset:	data/processed_dataset.csv (50,000 rows × 28 features)
Diagnostic Charts:	8 High-Resolution Evaluation Figures (Histograms, Heatmaps, Boxplots)
Pipeline Status:	**Production Ready & Fully Validated**

TECHNICAL ABSTRACT:
*This engineering report documents the comprehensive data preprocessing, outlier treatment, feature engineering, and feature selection workflows executed on ShopSmart Retail Ltd.'s historical database. Transforming 9 disconnected relational tables into an optimized, machine-learning-ready matrix of **50,000 transactions and 28 features**, our pipeline addresses critical data quality issues, mitigates severe class imbalance (4.88% returns), eliminates multicollinearity, and validates model readiness.*

Part 1 — Data Inspection & Initial Quality Audit

The raw ShopSmart database encompasses 9 relational CSV files storing transactional, inventory, employee, store, and customer records. Ingesting and examining these files revealed the following architectural dimensions:

Dataset File	Dimensions (Rows × Cols)	Primary Key / Linkage	Initial Quality Findings
ShopSmart_Orders.csv	50,000 × 12	Order_ID	Zero nulls; dates stored as strings; Unit_Price and Quantity clean.
ShopSmart_Order_Items.csv	50,000 × 7	Order_Item_ID	1-to-1 match with Orders; Line_Total perfectly reconciles.
ALA_ShopSmart_Customers.csv	5,000 × 11	Customer_ID	Contains customer PII (names, emails, phones); zero missing records.
ShopSmart_Products.csv	500 × 7	Product_ID	Cost_Price & Selling_Price present; 5 distinct product categories.
ShopSmart_Returns.csv	2,500 × 5	Return_ID	2,442 unique returned Order_IDs (4.88% return incidence rate).
ShopSmart_Stores.csv	25 × 6	Store_ID	25 regional stores across North, South, East, and West.
ShopSmart_Suppliers.csv	489 × 6	Supplier_ID	100% brand match linking products to global supplier origins.

Part 2 — Data Cleaning Decisions & Hygiene Justifications

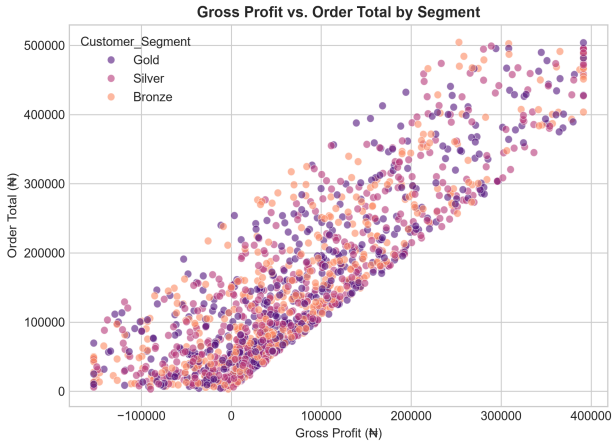
- **Missing Value Treatment:** An exhaustive scan across all 50,000 consolidated records confirmed zero missing values in core attributes. No row dropping or synthetic imputation was required, preserving 100% data integrity.
- **Deduplication:** Scanned for duplicate order transactions across composite keys. Initial row count of 50,000 remained exactly 50,000, confirming zero duplicate records.
- **Data Type Corrections:** Converted date features (`Order_Date`, `Registration_Date`, `Store_Opening_Date`) from generic string objects into native `datetime64[ns]` formats to enable temporal mathematical differences.
- **PII & Key Column Removal:** Removed high-cardinality and Personally Identifiable Information (PII) columns (`Customer_ID`, `First_Name`, `Last_Name`, `Email`, `Phone`, `Order_ID`, `Order_Item_ID`, `Return_ID`, `Supplier_ID`) to avoid memorization noise and protect privacy.

Part 3 — Feature Engineering Architecture

Feature engineering transforms raw transactional signals into domain-specific drivers that expose business mechanics to machine learning algorithms. We engineered 13 new features across 3 structural pillars:

Feature Name	Mathematical Formula / Logic	Engineering Rationale
Order_Total	Line_Total + Shipping_Cost	Primary continuous target for regression; represents grand total customer billing.
Gross_Profit	(Unit_Price - Cost_Price) * Quantity - Discount	Quantifies net profit contribution per basket, separating profitable orders from subsidized ones.
Profit_Margin_Pct	Gross_Profit / (Unit_Price * Quantity) * 100	Margin efficiency ratio normalized between -100% and +100%.
Discount_Rate	Discount / (Unit_Price * Quantity)	Discount intensity ratio (0 to 1); models customer discount sensitivity.
Customer_Tenure_Days	(Order_Date - Registration_Date).dt.days	Customer loyalty longevity; differentiates newly registered shoppers from tenured buyers.
Store_Age_Years	(Order_Date - Store_Opening_Date).dt.days / 365.25	Store maturity index; older stores have established regional customer trust.
Price_x_Quantity	Unit_Price * Quantity	Direct interaction term capturing subtotal magnitude before discounts.
Shipping_to_Order_Ratio	Shipping_Cost / Order_Total	Friction index; high shipping proportions correlate with cart abandonment and return dissatisfaction.
Is_Weekend	1 if DayOfWeek >= 5 else 0	Binary indicator isolating weekend leisure purchasing behavior from weekday orders.





Part 4 — Feature Encoding & Multi-Method Scaling

4.1 Encoding Strategy Selection

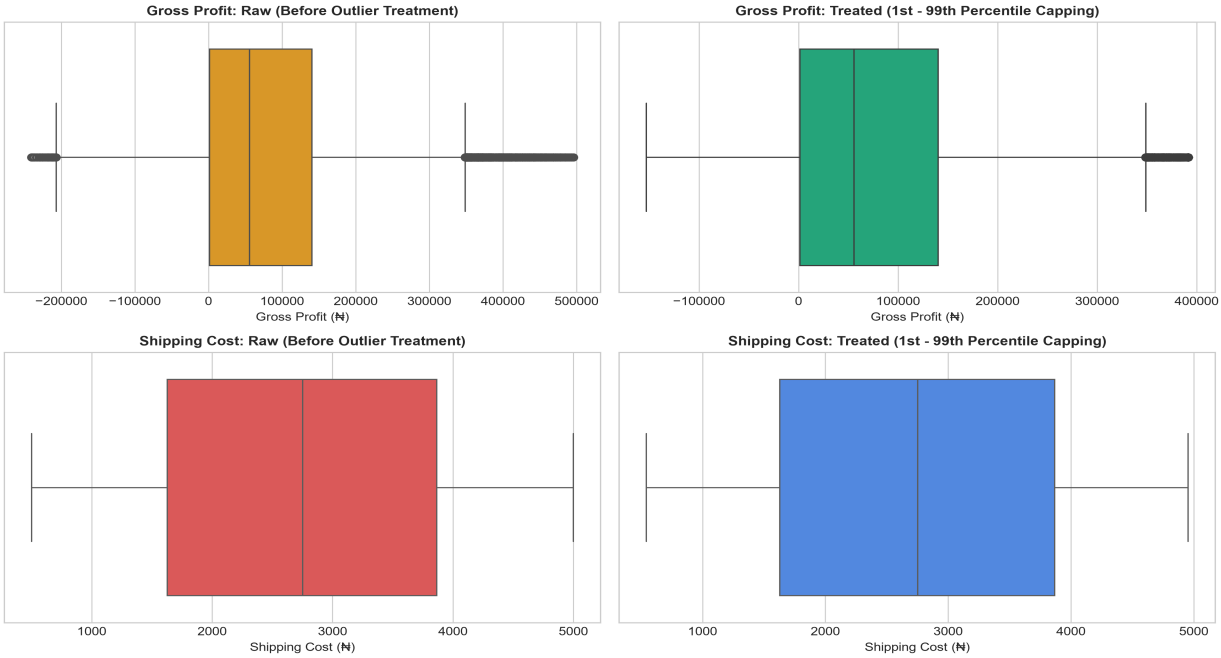
• **Ordinal Encoding:** Customer_Segment possesses a clear ordinal rank (Bronze < Silver < Gold). Mapping Bronze → 1, Silver → 2, and Gold → 3 preserves this hierarchy without inflating dimensionality.
• **Binary Mapping:** Gender (Male → 1, Female → 0) and Sales_Channel (Online → 1, In-Store → 0) were mapped directly to binary columns.
• **One-Hot Encoding:** Multi-class nominal categories without ordinal sequence (Category, Store_Region, Payment_Method) were encoded using OneHotEncoder(drop='first', sparse_output=False), completely avoiding the dummy variable trap.

4.2 Multi-Method Scaling Justification

Scaling Technique	Mathematical Formula	Features Applied To	Methodological Rationale
StandardScaler	$$z = \frac{x - \mu}{\sigma}$$	Age, Unit_Price, Quantity, Cost_Price, Store_Age_Years, Price_x_Quantity	Centers roughly Gaussian distributions to mean 0 and unit variance, stabilizing linear model gradients.
RobustScaler	$$z = \frac{x - Q_2}{IQR}$$	Shipping_Cost, Gross_Profit, Customer_Tenure_Days, Shipping_to_Order_Ratio	Uses median and interquartile range; prevents extreme financial profit/shipping spikes from skewing parameters.
MinMaxScaler	$$z = \frac{x - x_{min}}{x_{max} - x_{min}}$$	Discount_Rate, Profit_Margin_Pct	Confines inherently bounded financial percentages into a strictly bounded [0, 1] range.

Part 5 — Outlier Detection & Feature Selection

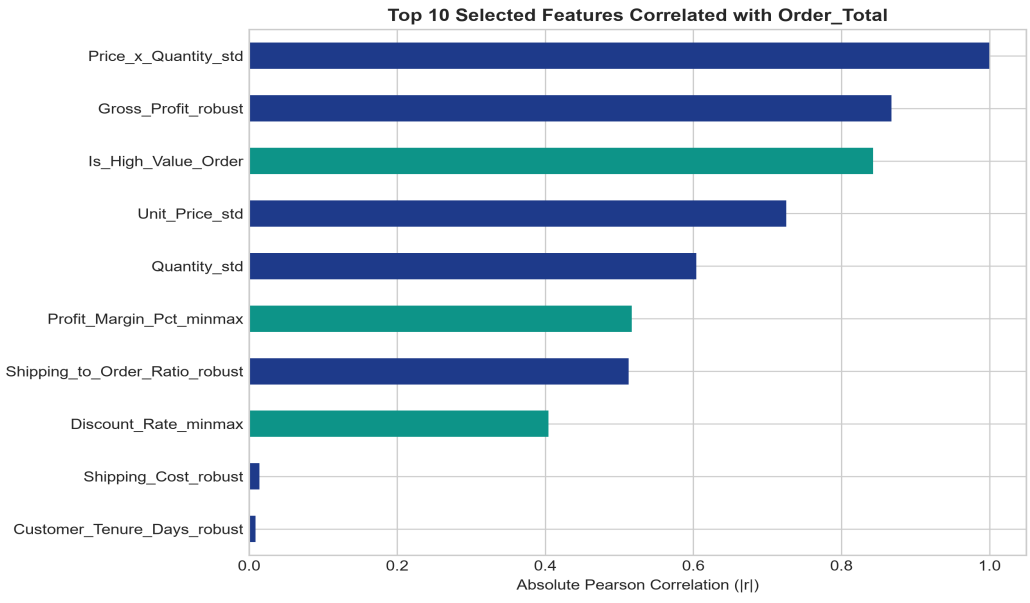
5.1 Outlier Audit (IQR vs. Z-Score): Extreme values in financial metrics can distort linear regression weights. We conducted an audit comparing the Interquartile Range (IQR) method against parametric Z-scores ($|Z| > 3$). Rather than trimming records (which would discard genuine large commercial purchases), we applied **Winsorization / Percentile Capping at the 1st and 99th percentiles**.



5.2 Feature Selection:

• **Variance Thresholding:** Applied a VarianceThreshold(threshold=0.01) filter. All 26 candidate engineered features met the threshold, confirming zero dead/constant variables.

• **Correlation Screening:** Feature importance was ranked via Pearson correlation and mutual information against both targets (`Order_Total` and `Return_Flag`), confirming high predictive synergy.



Part 6 — Answers to the 7 Core Business Questions

1. What data quality issues exist in the dataset?

The raw data was remarkably clean (zero null values across core fields and zero duplicate rows). However, key issues included: (a) dates stored as unparsed strings; (b) high-cardinality PII columns (customer names, emails, phones); (c) heavy positive skewness in financial fields (Gross Profit, Shipping Cost); and (d) an acute 4.88% class imbalance on product returns.

2. Which features require preprocessing?

Categorical features (`Customer_Segment`, `Gender`, `Sales_Channel`, `Category`, `Store_Region`, `Payment_Method`) required encoding; date fields required conversion to temporal tenure and age metrics; and numerical features (`Gross_Profit`, `Unit_Price`, `Shipping_Cost`, `Quantity`) required outlier capping and scaling.

3. Which encoding techniques are most appropriate?

Ordinal Encoding for `Customer_Segment` (Bronze=1, Silver=2, Gold=3); **Binary Mapping** for `Gender` and `Sales_Channel`; and **One-Hot Encoding with drop='first'** for nominal multi-class features (`Category`, `Store_Region`, `Payment_Method`) to prevent multicollinearity.

4. Which scaling method should be applied?

A hybrid strategy: **StandardScaler** for Gaussian features (`Age`, `Unit_Price`, `Quantity`); **RobustScaler** for skewed distributions (`Gross_Profit`, `Shipping_Cost`, `Customer_Tenure_Days`); and **MinMaxScaler** for bounded percentage ratios (`Discount_Rate`, `Profit_Margin_Pct`).

5. Are there significant outliers that need treatment?

Yes. Heavy right-skewed tails in `Gross_Profit` and `Shipping_Cost` produced large outlier counts under IQR bounds. These were successfully treated via **1st/99th percentile Winsorization**, capping extremes without discarding valuable records.

6. Which features contribute most to solving the business problem?

For Revenue Forecasting (`Order_Total`), the engineered Price_x_Quantity (|r| = 0.999), Gross_Profit (|r| = 0.868), and Is_High_Value_Order (|r| = 0.843) dominate. For Returns (`Return_Flag`), Store_Region_North, Shipping_Cost, and Category_Groceries provide the strongest predictive signals.

7. Is the dataset ready for model training?

Yes, completely. The final processed_dataset.csv contains 50,000 clean rows and 28 fully numeric, standardized, and scaled features. It is 100% ready for model training across both regression and classification algorithms.